9.3 Resistivity

Question Paper

Course	CIE A Level Physics
Section	9. Electricity
Topic	9.3 Resistivity
Difficulty	Medium

Time allowed: 40

Score: /29

Percentage: /100

Question 1a

An electrical heating element, made from uniform nichrome wire, is required to dissipate 600 W when connected to the 230 V mains supply.

The radius of the wire is 0.16 mm.

Calculate the length of nichrome wire required.

Resistivity of nichrome = 1.1×10^{-6} Ω m

[5 marks]

Question 1b

Suggest **two** properties that the nichrome wire must have to make it suitable as an electrical heating element.

[2 marks]

Question 1c

Explain why the resistivity of the nichrome wire changes with temperature.

[3 marks]

Question 1d

An engineer wants to use the same uniform nichrome wire to use in another identical electrical heating element which is also required to dissipate 600 W when connected to the 230 V mains supply.

The only nichrome wire they have available has a radius of 0.08 mm.

Calculate the length needed for this new nichrome wire to produce the same current through the heating element.

[3 marks]

Question 2a

A student investigates the resistivity ρ of a length of a wire l, with resistance R, diameter d and a potential difference V through it.

Determine the resistivity, in terms of ρ, of a wire of the same length with a diameter $2d$ and a potential difference $3V$ through it instead.

[3 marks]

Question 2b

The student decides to investigate how the cross-sectional area of a wire affects its resistance. The circuit they use to carry out the experiment is shown in Fig 1.1.

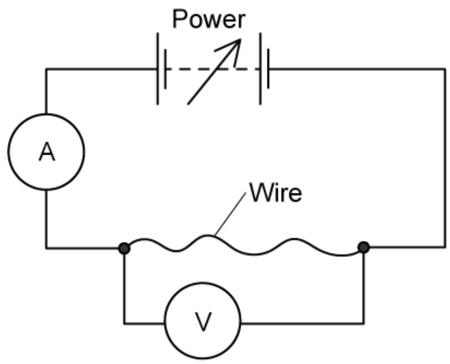


Fig 1.1

Describe a method to accurately determine the cross-sectional area of the wires.

[3 marks]

Question 2c

The student uses three pieces of nichrome wire, each 1.5 m long. The cross-sectional area of the pieces of wire are

$0.45\ m^2$ $0.21\ m^2$ $0.89\ m^2$

Sketch a graph of voltage *V* against the current *I* for each of the wires.

[3 marks]

Question 3a

At room temperature, a metal has a resistivity of 3.0×10^{-7} Ω m.

A 2.5 m length of wire made from this metal has a cross-sectional area of 0.50 mm^2.

Calculate the power dissipated in this length of wire when it carries a current of 10 mA.

[4 marks]

Question 3b

State and explain the assumption that is necessary for calculating the answer to part (a).

[2 marks]

Question 3c

In electrical power transmission conventional copper wires are often replaced with wires which conduct with wires with an effective resistance which is close to zero, called superconducting wires.

Explain why the superconducting wires increase the efficiency of electrical power transmission.

[1 mark]

Model Answers: Medium

1a

(a) Calculate the length of nichrome wire required:

List the known quantities:

- Radius of the wire, $r = 0.16$ mm $= 0.16 \times 10^{-3}$ m
- Potential difference through wire, $V = 230$ V
- Power through wire, $P = 600$ W
- Resistivity of nichrome, $\rho = 1.1 \times 10^{-6}\ \Omega$ m

Use the equation relating power and resistance and calculate to find resistance, R:

- $P = \frac{V^2}{R}$
- $R = \frac{V^2}{P} = \frac{230^2}{600}$ [1 mark]
- $R = 88.2\ \Omega$ [1 mark]

Calculate cross-sectional area A:

- Area of a circle $A = \pi r^2 = \pi \times (0.16 \times 10^{-3})^2 = 8.042 \times 10^{-8}\ \text{m}^2$ [1 mark]

Use the resistivity equation to calculate length:

- $\rho = \frac{R\ A}{L}$
- $L = \frac{R\ A}{\rho} = \frac{88.2 \times (8.042 \times 10^{-8})}{(1.1 \times 10^{-6})}$ [1 mark]
- $L = 6.4\ \text{m}\ (2\ \text{s.f.})$ [1 mark]

[Total: 5 marks]

1b

(b) Two properties that the nichrome wire must have to make it suitable as an electrical heating element are:

Any **two** from:

- High resistance / resistivity / low electrical conductivity (so it gets hot); [1 mark]
- High melting point (to withstand high temperatures); [1 mark]
- Low thermal capacity / low specific heat capacity (to reduce the amount of energy needed to make it hot); [1 mark]

[Total: 2 marks]

1c

(c) The resistivity of the nichrome wire changes with temperature because:

- The resistivity / resistance increases with increasing temperature; [1 mark]
- This is due to the (lattice) ions / atoms vibrating more / with a greater amplitude; [1 mark]
- The rate of momentum of the charge carriers / electrons (along the wire) is therefore reduced; [1 mark]

[Total: 3 marks]

1d

(d) Calculate the length needed for this new nichrome wire to produce the same current through the heating element:

List the known quantities:

- Length of old wire $L = 6.4$ m (from part c)
- Old wire radius = 0.16 mm
- New wire radius = 0.08 mm
- Resistivity of nichrome, $\rho = 1.1 \times 10^{-6}\ \Omega$ m

Determine how the area would change for ½ r:

- Area of a circle, $A = \pi r^2$
- Therefore for $r' = ½ r$, $A' = \pi \left(\frac{r}{2}\right)^2 = \frac{\pi r^2}{4} = \frac{A}{4}$ **OR** new area is one quarter original area; [1 mark]

Determine how the length would change for ¼ A:

- $L = \frac{RA}{\rho}$
- $L' = \frac{R}{\rho} \times \frac{A}{4} = \frac{L}{4}$ **OR** new length is one-quarter original length; [1 mark]
- $L' = \frac{6.4}{4} = 1.6 \text{ m}$ [1 mark]

[Total: 3 marks]

2a

(a) The resistivity, in terms of ρ, of a wire with a diameter $2d$ and a potential difference $3V$ through it is:

Determine the equation for the resistivity of the first wire:

- $R = \frac{\rho l}{A} \Rightarrow \rho = \frac{RA}{l}$
- $\rho = \frac{R \times \pi r^2}{l} = \frac{R \times \pi \left(\frac{d}{2}\right)^2}{l}$ [1 mark]

Determine the new resistance, R:

- $R = \frac{V}{I}$ therefore $\frac{3V}{I} = 3R$ [1 mark]

Determine the new resistivity:

- $\rho_2 = \frac{3R \times \pi \left(\frac{2d}{2}\right)^2}{l} = \frac{(3 \times 2^2)R \times \pi \left(\frac{d}{2}\right)^2}{l}$

- $\rho_2 = \dfrac{12R \times \pi\left(\dfrac{d}{2}\right)^2}{l} = 12\rho$ [1 mark]

[Total: 3 marks]

2b

(b) The student would accurately determine the cross-sectional area of the wires by:

- Measuring the diameter at (at least) 3 different parts of each wire; [1 mark]
- Using a micrometer (screw gauge); [1 mark]
- Calculate an average diameter *d* and calculate the cross-sectional area *A* using $A = \pi\left(\dfrac{d}{2}\right)^2 = \dfrac{\pi d^2}{4}$ [1 mark]

[Total: 3 marks]

Make sure not to just stop your answer by measuring the diameter. The question asks how to measure the **area**, which requires a calculation using the diameter which you must provide to complete your answer.

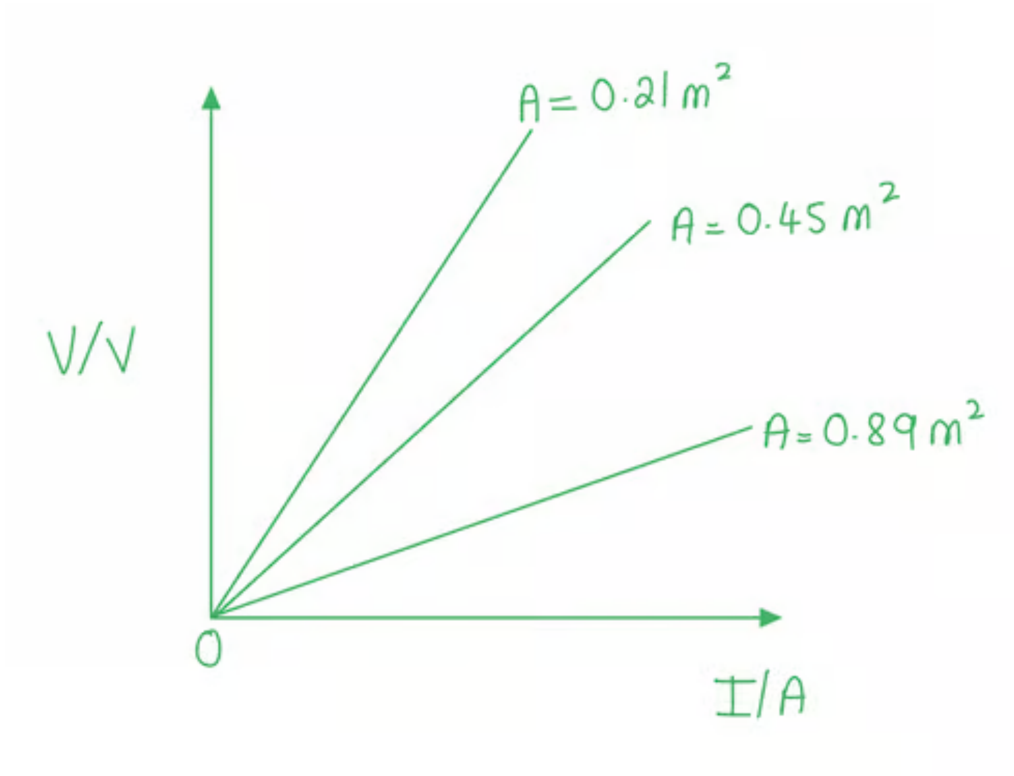
2c

(c) Sketch a graph of voltage *V* against the current *I* for each of the wires.

Determine how the cross-sectional area *A* related to the gradient of the graph:

- $R = \dfrac{V}{I}$ is the gradient of the graph
- $R = \dfrac{\rho l}{A}$ therefore $R \propto \dfrac{1}{A}$

A graph that would score 2 marks should look like:



- Voltage (V)/V labelled on the y-axis and current (I)/A labelled on the x-axis; [1 mark]
- Three straight lines drawn with a positive gradient; [1 mark]
- Labelled 0.21 m^2, 0.45 m^2 and 0.89 m^2 from steepest to shallowest gradient; [1 mark]

[Total: 3 marks]

You must be as accurate and neat as possible with any graph sketches. Always label the axes and add units where applicable!

3a

(a) Calculate the power dissipated in this length of wire when it carries a current of 10 mA:

List the known quantities:

- Resistivity of wire (at room temperature), $\rho = 3.0 \times 10^{-7}$ Ω m
- Cross-sectional area, A = 0.50 mm^2
- Length of the wire, L = 2.5 m
- Current, I = 10 mA = 10 × 10^{-3} A

- Current, I = 10 mA = 10 × 10^{-3} A

Convert the cross-section area into m^2:

- 1 mm = 1 × 10^{-3} m therefore 1 mm^2 = 1 × 10^{-6} m
- A = 0.50 mm^2 = 0.50 × 10^{-6} m^2 [1 mark]

Calculate the resistance of the wire at room temperature:

- $\rho = \frac{RA}{L}$
- $R = \frac{\rho L}{A} = \frac{(3.0 \times 10^{-7}) \times 2.5}{(0.5 \times 10^{-6})}$ [1 mark]
- $R = 1.5\ \Omega$ [1 mark]

Calculate the power dissipated:

- $P = I^2 R = (10 \times 10^{-3})^2 \times 1.5$
- $P = 1.5 \times 10^{-4}$ W [1 mark]

[Total: 4 marks]

3b

(b) Explain the assumption that is necessary for calculating the answer to part (a):

State the assumption:

- The resistance of the wire is constant; [1 mark]

Explain why the resistance must be assumed constant:

- If the resistance changes (because the temperature of the wire increases) then the current (hence power) will change; [1 mark]

[Total: 2 marks]

3c

(c) Explain why the superconducting wires increase the efficiency of electrical power transmission:

- If the resistance is zero, then there is no power loss / the waste energy transferred as heat is zero; [1 mark]

[Total: 1 mark]